



PT. SALIPI NAGARI PERKASA

Komp. Pejaten Mas Blok E No. 3 Kramat Watu Serang - Banten 42161

Telp. : 0254 - 231363 Fax.: 0254 - 231183

LAPORAN UJI MINYAK TRAFO

di

UNIT JASA PEMBANGKIT

JERANJANG



Yang dilakukan pada tanggal 20 JUNI 2019

Untuk keadaan trafo bulan JUNI 2019

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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFU UNIT 1	100691	4	OLTT+ and Corrosive Sulfur
2.	TRAFU 1 ESP UNIT 1	10937	5	OLTT+ and Corrosive Sulfur
3.	TRAFU 2 ESP UNIT 1	10035	6	OLTT+ and Corrosive Sulfur
4.	TRAFU 3 ESP UNIT 1	10031	7	OLTT+ and Corrosive Sulfur
5.	TRAFU 4 ESP UNIT 1	10034	8	OLTT+ and Corrosive Sulfur
6.	MAIN TRAFU UNIT 3	3011160192	23	OLTT+ and Corrosive Sulfur
7.	UAT 3	PX18-0082	22	OLTT+ and Corrosive Sulfur
8.	TRAFU 1 ESP UNIT 3	0073084208	11	OLTT+ and Corrosive Sulfur
9.	TRAFU 2 ESP UNIT 3	0070084208	12	OLTT+ and Corrosive Sulfur
10.	TRAFU 3 ESP UNIT 3	0072084208	13	OLTT+ and Corrosive Sulfur
11.	TRAFU 4 UNIT 3	0071084208	14	OLTT+ and Corrosive Sulfur
12.	ASH HANDLING-1	10820108	15	OLTT+ and Corrosive Sulfur
13.	ASH HANDLING-2	10820107	16	OLTT+ and Corrosive Sulfur
14.	TRAFU 1 ESP UNIT 2	10037	17	OLTT+ and Corrosive Sulfur
15.	TRAFU 2 ESP UNIT 2	10036	18	OLTT+ and Corrosive Sulfur
16.	TRAFU 3 ESP UNIT 2	10033	19	OLTT+ and Corrosive Sulfur
17.	TRAFU 4 ESP UNIT 2	10032	20	OLTT+ and Corrosive Sulfur
18.	MAIN TRAFU UNIT 2	1006902	21	OLTT+ and Corrosive Sulfur

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Note:

- * OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

- ** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAFO UNIT 1	100691	4

A) Analysis

- **Acidity** of 0.017mgKOH/g and **IFT** of 41.3mN/m are **ACCEPTABLE**.
- **Dielectric** of 66kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** shows combustible gasses levels to have stabilized. Proceed with annual monitoring.
- **Water Content** in oil of 7ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.38% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.24% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

4

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
22/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE	50	55				
24/07/2018	ONLINE					GOOD	
25/08/2018	OFFLINE						
25/09/2018	ONLINE						
23/10/2018	ONLINE						
22/11/2018	ONLINE						
08/12/2018	ONLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
22/05/2018	ROUTINE	0.011 AC		77 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.014 AC	41.9 AC	69 AC	0.5 AC	0.8716 AC	CLEAR AC	
24/07/2018	ROUTINE	0.014 AC		86 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.017 AC		90 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.014 AC		96 AC	0.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.014 AC		83 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.014 AC		73 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.014 AC		75 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.014 AC		71 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.017 AC	41.3 AC	66 AC	0.5 AC	0.8789 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	147	9471	43278	32	655	2728	4	5	ND	843	56320
06/12/2017	183	12895	43794	39	715	3166	5	5	ND	947	60802
28/06/2018	176	11044	46776	41	758	3393	5	6	ND	986	62199
20/06/2019	18	19122	60738	26	756	3272	2	3	ND	805	83937

RECOMMENDATION:

GAS LEVELS HAVE REMAINED STABLE. PROCEED WITH ANNUAL MONITORING.

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
22/05/2018	30	10	1.05	B
28/06/2018	41	7	0.35	A
24/07/2018	32	6	0.57	A
25/08/2018	30	6	0.63	A
25/09/2018	38	7	0.46	A
23/10/2018	44	8	0.36	A
22/11/2018	41	9	0.45	A
08/12/2018	41	8	0.40	A
21/05/2019	38	8	0.52	A
20/06/2019	39	7	0.38	A

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

4

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.02	0.01	0.01
06/12/2017	ND	0.01	0.04
28/06/2018	ND	ND	0.04
20/06/2019	0.01	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.140 AC
06/12/2017	0.000 AC	0.040 AC
28/06/2018	0.000 AC	0.140 AC
20/06/2019	0.000 AC	0.180 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.24 AC
06/12/2017	0.27 AC
28/06/2018	0.24 AC
20/06/2019	0.24 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.017mgKOH/g and **IFT** of 38.7mN/m are **ACCEPTABLE**.
- **Dielectric** of 34kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 23ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 1.50% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates high moisture content in the oil.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected were less than 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur test** on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- Retest **Water Content** and **Dielectric** in 6 months to assess if corrective work on the transformer oil is required.
- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes.**

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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	ONLINE						
23/10/2018	ONLINE						
22/11/2018	ONLINE						
08/12/2018	OFFLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.065 QU		38 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.067 QU	25.9 UN	32 AC	1.5 AC	0.8560 AC	CLEAR AC	
24/07/2018	ROUTINE	0.073 QU		38 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.020 AC		71 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.017 AC		63 AC	0.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.020 AC		49 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.017 AC		30 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.017 AC		41 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.017 AC		40 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.017 AC	38.7 AC	34 AC	0.5 AC	0.8747 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	20494	67487	4	69	1143	ND	1	ND	109	89233
06/12/2017	39	19933	67198	4	83	1261	ND	1	ND	127	88519
28/06/2018	44	20039	64529	4	119	1696	ND	2	ND	169	86433
20/06/2019	22	27043	68613	5	36	689	ND	ND	ND	63	96408

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	33	3.13	E
28/06/2018	32	30	2.85	E
24/07/2018	29	19	2.00	D
25/08/2018	32	12	1.14	B
25/09/2018	35	18	1.28	B
23/10/2018	40	24	1.32	B
22/11/2018	37	23	1.50	C
08/12/2018	32	18	1.71	D
21/05/2019	39	22	1.21	B
20/06/2019	38	23	1.50	C

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

5

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	95	ND	ND	ND	95
06/12/2017	ND	75	ND	ND	ND	75
28/06/2018	ND	85	ND	ND	ND	85
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.03
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.870 AC
06/12/2017	0.000 AC	0.570 AC
28/06/2018	0.000 AC	0.680 AC
20/06/2019	0.000 AC	0.230 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	ND UN
28/06/2018	ND UN
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	1A	LIGHT ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.020mgKOH/g is **ACCEPTABLE** while **IFT** of 27.9mN/m is **UNACCEPTABLE**. Although the level remained stable, the oil are containing oxidation by-products. These oxidation by-products are depositing and degrading the oil as well as the paper insulation. Therefore, removal of these contaminations are highly recommended.
- **Dielectric** of 68kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 17ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.11% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

Page Disclaimer :

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts as well as moisture from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

6

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			LOW		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.056 QU		45 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.059 QU	27.6 UN	55 AC	1.5 AC	0.8570 AC	CLEAR AC	
24/07/2018	ROUTINE	0.062 QU		47 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.022 AC		56 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.020 AC		67 AC	0.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.020 AC		71 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.020 AC		63 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.020 AC		51 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.020 AC		71 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.020 AC	27.9 UN	68 AC	0.5 AC	0.8658 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	27	18639	66781	3	64	1106	ND	1	ND	95	86621
06/12/2017	20	19360	61590	3	45	728	ND	1	ND	69	81747
28/06/2018	22	22494	64443	4	57	789	ND	ND	ND	83	87809
20/06/2019	23	17710	61707	5	23	842	ND	ND	ND	51	80310

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	20	1.90	D
28/06/2018	32	17	1.62	C
24/07/2018	30	15	1.58	C
25/08/2018	31	13	1.24	B
25/09/2018	31	11	1.04	B
23/10/2018	33	14	1.15	B
22/11/2018	34	18	1.48	C
08/12/2018	33	14	1.15	B
21/05/2019	37	15	0.98	B
20/06/2019	37	17	1.11	B

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

6

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	62	ND	ND	ND	62
06/12/2017	ND	67	ND	ND	ND	67
28/06/2018	ND	84	ND	ND	ND	84
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	0.02
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.900 AC
06/12/2017	0.000 AC	0.980 AC
28/06/2018	0.000 AC	1.310 AC
20/06/2019	0.000 AC	0.680 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.02 UN
06/12/2017	0.02 UN
28/06/2018	ND UN
20/06/2019	ND UN

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	1B	DARK ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.051mgKOH/gm and **IFT** of 29.6mN/m are **QUESTIONABLE**.
Although the level remained stable, the oil are containing oxidation by-products. These oxidation by-products are depositing and degrading the oil as well as the paper insulation. Therefore, removal of these contaminations are highly recommended.
- **Dielectric** of 30kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 19ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.35% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

7

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
23/10/2018	ONLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.056 QU		48 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.053 QU	29.7 QU	62 AC	1.5 AC	0.8605 AC	CLEAR AC	
24/07/2018	ROUTINE	0.051 QU		64 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.056 QU		57 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.051 QU		46 AC	1.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.056 QU		44 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.056 QU		39 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.056 QU		39 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.051 QU		33 AC	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.051 QU	29.6 QU	30 AC	1.5 AC	0.8689 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	32	15273	65111	5	95	1002	ND	2	ND	134	81520
06/12/2017	7	14730	65738	3	42	617	ND	ND	ND	52	81137
28/06/2018	9	17958	66007	4	46	781	ND	1	ND	60	84806
20/06/2019	16	17771	65478	6	73	829	ND	1	ND	96	84174

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	21	2.00	D
28/06/2018	30	17	1.78	D
24/07/2018	30	17	1.78	D
25/08/2018	30	15	1.58	C
25/09/2018	32	21	2.00	D
23/10/2018	36	20	1.42	C
22/11/2018	36	22	1.56	C
08/12/2018	35	20	1.42	C
21/05/2019	36	19	1.35	B
20/06/2019	36	19	1.35	B

FIELD SERVICE

DATE SERVICE

DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	32	ND	ND	ND	32
06/12/2017	ND	14	ND	ND	ND	14
28/06/2018	ND	20	ND	ND	ND	20
20/06/2019	ND	57	ND	ND	ND	57

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.11
20/06/2019	0.01	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.700 AC
06/12/2017	0.000 AC	0.630 AC
28/06/2018	0.000 AC	1.030 AC
20/06/2019	0.000 AC	0.520 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.01 UN
28/06/2018	ND UN
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 38.6mN/m are **ACCEPTABLE**.
- **Dielectric** of 45kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 19ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.24% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factor** for 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes.**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

8

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			LOW	FAIR	GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
23/10/2018	ONLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.034 AC		37 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.037 AC	27.5 UN	55 AC	1.5 AC	0.8635 AC	CLEAR AC	
24/07/2018	ROUTINE	0.039 AC		50 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.017 AC		61 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.014 AC		45 AC	0.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.014 AC		52 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.014 AC		44 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.014 AC		47 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.014 AC		41 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.014 AC	38.6 AC	45 AC	0.5 AC	0.8775 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	13	12265	66770	5	172	1206	ND	8	ND	198	80439
06/12/2017	8	13331	63527	3	70	720	ND	1	ND	82	77660
28/06/2018	9	17962	64248	4	75	1054	ND	2	ND	90	83354
20/06/2019	11	15056	64828	5	11	951	ND	ND	ND	27	80862

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	23	2.18	D
28/06/2018	30	20	2.10	D
24/07/2018	31	19	1.80	D
25/08/2018	30	14	1.47	C
25/09/2018	30	20	2.10	D
23/10/2018	41	21	1.05	B
22/11/2018	38	21	1.37	C
08/12/2018	38	20	1.30	B
21/05/2019	36	19	1.35	B
20/06/2019	37	19	1.24	B

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

8

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	18	ND	ND	ND	18
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	21	ND	ND	ND	21
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.05	0.02	0.01
06/12/2017	ND	ND	0.02
28/06/2018	ND	ND	0.05
20/06/2019	ND	ND	0.02

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

LIQUID POWER FACTOR (ASTM D924)

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	25°C	100°C
04/07/2017	0.180 QU	6.680 UN
06/12/2017	0.010 AC	2.180 AC
28/06/2018	0.060 AC	2.850 AC
20/06/2019	0.010 AC	1.120 AC

DATE	%BY WEIGHT
04/07/2017	0.08 UN
06/12/2017	0.06 UN
28/06/2018	0.02 UN
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA

DATE	PPM
------	-----

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
6.	MAIN TRAF0 UNIT 3	3011160192	23

A) Analysis

- **Acidity** of 0.011mgKOH/g and **IFT** of 40.6mN/m are **ACCEPTABLE**.
- **Dielectric** of 83kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 8ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.52% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

Page Disclaimer :

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes.**

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- PAUWES TRAF0
DATE MFG	- 2018
MAIN TANK S/N	- 3011160192
KVA	- 60000
PRI	- 157000 V
SEC	- 6300 V
KILOGRAMS	- 103500
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.171%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30183

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2019	ROUTINE	0.011 AC		93 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.011 AC	40.6 AC	83 AC	0.5 AC	0.8786 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
20/06/2019	3	9903	31346	6	156	369	ND	ND	ND	165	41783

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2019	38	7	0.46	A
20/06/2019	38	8	0.52	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 3

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- PAUWES TRAFO
DATE MFG	- 2018
MAIN TANK S/N	- 3011160192
KVA	- 60000
PRI	- 157000 V
SEC	- 6300 V
KILOGRAMS	- 103500
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.171%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30183
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
20/06/2019	0.000 AC	0.090 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
20/06/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
20/06/2019	

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
7.	UAT 3	PX18-0082	22

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 37.0mN/m are **ACCEPTABLE**.
- **Dielectric** of 89kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 5ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.41% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes.**

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - UAT 3

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- B & D
DATE MFG	- 2018
MAIN TANK S/N	- PX18-0082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 52000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 12000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2019	ROUTINE	0.014 AC		85 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.014 AC	37.0 AC	89 AC	0.5 AC	0.8798 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
20/06/2019	4	10251	34223	16	39	232	26	2	ND	87	44793

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2019	35	5	0.35	A
20/06/2019	34	5	0.41	A

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **UAT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- B & D
DATE MFG	- 2018
MAIN TANK S/N	- PX18-0082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 52000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 12000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
20/06/2019	0.000 AC	0.120 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
------	--------	-------	------------

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	0073084208	11

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 35.2mN/m are **ACCEPTABLE**.
- **Dielectric** of 65kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 9ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.17% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.16% by weight is **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

B) Recommendation

- Monitor the **Specific Gravity** of the oil every 12 months.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
23/10/2018	ONLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.022 AC		31 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.022 AC	35.0 AC	60 AC	0.5 AC	0.8154 QU	CLEAR AC	
24/07/2018	ROUTINE	0.020 AC		53 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.022 AC		77 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.020 AC		64 AC	0.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.020 AC		63 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.020 AC		53 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.020 AC		62 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.022 AC		85 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.022 AC	35.2 AC	65 AC	0.5 AC	0.8212 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	14	12943	68807	3	62	1243	ND	ND	ND	79	83072
06/12/2017	14	12739	65097	3	66	612	ND	ND	ND	83	78531
28/06/2018	19	15552	64448	3	57	1391	ND	ND	ND	79	81470
20/06/2019	11	19448	64974	6	15	1064	ND	ND	ND	32	85518

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	27	2.57	D
28/06/2018	32	10	0.95	B
24/07/2018	26	9	1.17	B
25/08/2018	26	9	1.17	B
25/09/2018	30	10	1.05	B
23/10/2018	31	12	1.14	B
22/11/2018	32	9	0.86	B
08/12/2018	29	9	0.94	B
21/05/2019	25	9	1.17	B
20/06/2019	26	9	1.17	B

FIELD SERVICE

DATE	SERVICE
------	---------

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

11

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	ND
06/12/2017	0.01	ND	ND
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.350 AC
06/12/2017	0.000 AC	0.040 AC
28/06/2018	0.000 AC	0.050 AC
20/06/2019	0.000 AC	0.060 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.18 QU
28/06/2018	0.16 QU
20/06/2019	0.16 QU

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2E	BRASSY GOLD
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.28mgKOH/g is **ACCEPTABLE** while **IFT** of 31.3mN/m is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 41kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 18ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.71% M/dw is graded as **D**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. None of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.19% by weight is **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

Page Disclaimer :

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B) Recommendation

- Monitor the **Specific Gravity** of the oil every 12 months.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						
24/07/2018	ONLINE						
25/08/2018	OFFLINE						
25/09/2018	ONLINE						
23/10/2018	OFFLINE						
22/11/2018	ONLINE						
08/12/2018	ONLINE						
21/05/2019	OFFLINE						
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.022 AC		28 QU	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	30.2 QU	34 AC	0.5 AC	0.8080 QU	CLEAR AC	
24/07/2018	ROUTINE	0.022 AC		30 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.025 AC		67 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.022 AC		31 AC	0.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.025 AC		54 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.025 AC		33 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.025 AC		30 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.028 AC		40 AC	1.0 AC		CLEAR AC	
20/06/2019	ROUTINE	0.028 AC	31.3 QU	41 AC	1.0 AC	0.8163 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	38	16781	70523	5	129	2431	3	5	ND	180	89915
06/12/2017	42	19523	66059	3	102	912	ND	ND	ND	147	86641
28/06/2018	43	22272	68719	4	38	1226	ND	ND	ND	85	92302
20/06/2019	13	21221	65168	7	48	1870	ND	ND	ND	68	88327

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	30	2.85	E
28/06/2018	30	16	1.68	C
24/07/2018	32	25	2.38	D
25/08/2018	26	11	1.43	C
25/09/2018	39	32	1.76	D
23/10/2018	32	13	1.24	B
22/11/2018	42	38	1.90	D
08/12/2018	43	32	1.44	C
21/05/2019	33	18	1.48	C
20/06/2019	32	18	1.71	D

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

12

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	169	ND	ND	169
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	180	ND	ND	180

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.330 AC
06/12/2017	0.000 AC	0.030 AC
28/06/2018	0.000 AC	0.040 AC
20/06/2019	0.000 AC	0.010 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.20 AC
28/06/2018	0.19 QU
20/06/2019	0.19 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	3A	MAGENTA
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.020mgKOH/g and **IFT** of 38.5mN/m are **ACCEPTABLE**.
- **Dielectric** of 36kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 23ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 1.63% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm. This indicates higher than normal moisture content in the oil.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.19% by weight is **QUESTIONABLE**.
Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

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B) Recommendation

- Monitor the **Specific Gravity** of the oil every 12 months.
- Retest **Moisture** and **Dielectric** in 6 months to assess if corrective work on the transformer oil is required.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	ONLINE						
25/08/2018	OFFLINE						
25/09/2018	ONLINE						
23/10/2018	ONLINE						
22/11/2018	ONLINE						
08/12/2018	ONLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.014 AC		54 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.017 AC	38.4 AC	43 AC	0.5 AC	0.8075 QU	CLEAR AC	
24/07/2018	ROUTINE	0.014 AC		43 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.014 AC		76 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.020 AC		47 AC	0.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.014 AC		49 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.017 AC		39 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.014 AC		34 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.014 AC		41 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.020 AC	38.5 AC	36 AC	0.5 AC	0.8215 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	21	16364	70694	3	38	1068	ND	ND	ND	62	88188
06/12/2017	15	14515	65277	2	40	559	ND	ND	ND	57	80408
28/06/2018	30	15202	64188	4	65	1833	ND	ND	ND	99	81322
20/06/2019	14	18302	64811	6	86	2181	ND	ND	ND	106	85400

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	32	23	2.18	D
28/06/2018	35	22	1.56	C
24/07/2018	34	20	1.64	C
25/08/2018	25	11	1.43	C
25/09/2018	40	20	1.10	B
23/10/2018	39	15	0.82	A
22/11/2018	40	15	0.82	A
08/12/2018	43	18	0.81	A
21/05/2019	36	20	1.42	C
20/06/2019	35	23	1.63	C

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

13

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	4	ND	ND	ND	4
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.02
20/06/2019	0.01	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.620 AC
06/12/2017	0.000 AC	0.080 AC
28/06/2018	0.000 AC	0.010 AC
20/06/2019	0.000 AC	0.070 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.21 AC
28/06/2018	0.20 AC
20/06/2019	0.19 QU

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2E	BRASSY GOLD
28/06/2018	NON-COR	2E	BRASSY GOLD
20/06/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.022mgKOH/g is **ACCEPTABLE** while **IFT** of 30.3mN/m is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 51kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 12ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.42% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in dark orange, thus **Non-corrosive of 1B**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

Page Disclaimer :

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
06/12/2017	OFFLINE						
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	ONLINE						
23/10/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.028 AC		37 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	30.5 QU	71 AC	1.0 AC	0.8442 AC	CLEAR AC	
24/07/2018	ROUTINE	0.022 AC		42 AC	1.0 AC		CLEAR AC	
25/08/2018	ROUTINE	0.025 AC		66 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.022 AC		51 AC	1.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.022 AC		56 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.022 AC		44 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.022 AC		49 AC	1.0 AC		CLEAR AC	
21/05/2019	ROUTINE	0.022 AC		58 AC	1.0 AC		CLEAR AC	
20/06/2019	ROUTINE	0.022 AC	30.3 QU	51 AC	1.0 AC	0.8451 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	10	18669	66036	3	24	665	ND	ND	ND	37	85407
06/12/2017	14	15414	63712	2	22	675	ND	ND	ND	38	79839
28/06/2018	20	14690	67822	2	22	683	ND	ND	ND	44	83239
20/06/2019	19	16312	66417	4	37	732	ND	ND	ND	60	83521

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	30	21	2.20	D
28/06/2018	36	11	0.78	A
24/07/2018	26	11	1.43	C
25/08/2018	27	11	1.30	B
25/09/2018	30	11	1.15	B
23/10/2018	30	12	1.26	B
22/11/2018	30	11	1.15	B
08/12/2018	29	13	1.37	C
21/05/2019	26	11	1.43	C
20/06/2019	27	12	1.42	C

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

14

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	ND	ND
06/12/2017	ND	0.03	ND
28/06/2018	ND	0.03	0.02
20/06/2019	ND	0.03	0.01

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.360 AC
06/12/2017	0.000 AC	0.370 AC
28/06/2018	0.000 AC	0.550 AC
20/06/2019	0.000 AC	0.490 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.03 UN
28/06/2018	ND UN
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1B	DARK ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	1B	DARK ORANGE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
12.	ASH HANDLING-1	10820108	15

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 36.6mN/m are **ACCEPTABLE**.
- **Dielectric** of 27kV is **QUESTIONABLE**. A low value for the dielectric-breakdown voltage generally serves to indicate the presence of contaminants such as water, dirt, or other conducting particles in the oil.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** shows Carbon Monoxide and Carbon Dioxide levels to have stabilized. Other gases remain within normal limits and does not indicate any fault afflicting the transformer unit. Proceed with annual monitoring.
- **Water Content** in oil of 29ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.38% M/dw is graded as **C**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.21% by weight is detected, thus **ACCEPTABLE**.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

Page Disclaimer :

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B) Recommendation

- Retest **Water Content** and **Dielectric** in 6 months to assess if corrective work on the transformer oil is required.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes.**

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - ASH HANDLING-1

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
06/12/2017	ONLINE	52					
22/05/2018	ONLINE	45		NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	ONLINE						
25/08/2018	OFFLINE						
23/10/2018	ONLINE						
22/11/2018	ONLINE						
08/12/2018	ONLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
22/05/2018	ROUTINE	0.022 AC		33 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	37.5 AC	42 AC	1.0 AC	0.8804 AC	CLEAR AC	
24/07/2018	ROUTINE	0.022 AC		39 AC	1.0 AC		CLEAR AC	
25/08/2018	ROUTINE	0.025 AC		60 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.025 AC		43 AC	1.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.022 AC		28 QU	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.022 AC		25 QU	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.022 AC		20 UN	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.022 AC		28 QU	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.022 AC	36.6 AC	27 QU	1.5 AC	0.8890 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	10984	71058	20	441	5037	7	8	ND	511	87590
06/12/2017	49	12034	67124	20	558	5440	8	8	ND	643	85241
28/06/2018	53	10939	69394	20	557	5425	9	8	ND	647	86405
20/06/2019	53	14032	65585	13	487	5360	ND	11	ND	564	85541

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
GAS LEVELS HAVE REMAINED STABLE. PROCEED WITH ANNUAL MONITORING.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
22/05/2018	33	26	2.13	C
28/06/2018	32	24	2.28	C
24/07/2018	32	20	1.90	B
25/08/2018	30	16	1.68	B
25/09/2018	35	17	1.21	A
23/10/2018	36	29	2.06	C
22/11/2018	36	28	1.99	B
08/12/2018	35	28	1.99	B
21/05/2019	34	29	2.38	C
20/06/2019	34	29	2.38	C

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	0.01
28/06/2018	ND	0.01	0.04
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.730 AC
06/12/2017	0.000 AC	0.240 AC
28/06/2018	0.000 AC	0.460 AC
20/06/2019	0.000 AC	0.360 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC
06/12/2017	0.24 AC
28/06/2018	0.22 AC
20/06/2019	0.21 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
13.	ASH HANDLING-2	10820107	16

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 36.8mN/m are **ACCEPTABLE**.
- **Dielectric** of 37kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** shows Carbon Monoxide and Carbon Dioxide gas levels is high. However, the Gas Generation Rate is within acceptable limit. Other gasses are assessed as acceptable and did not indicate any fault afflicting the transformer unit. For this case, may proceed with annual monitoring.
- **Water Content** in oil of 24ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.97% M/dw is graded as **B**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.20% by weight is detected, thus **ACCEPTABLE**.
- **Corrosive sulfur** test on copper strip resultant in magenta, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - ASH HANDLING-2

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
27/11/2017	OFFLINE	40					
06/12/2017	ONLINE	48					
22/05/2018	ONLINE	45				GOOD	NONE
28/06/2018	OFFLINE						
24/07/2018	ONLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
08/12/2018	ONLINE						
21/05/2019	OFFLINE						
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
22/05/2018	ROUTINE	0.020 AC		24 UN	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.022 AC	35.0 AC	50 AC	1.5 AC	0.8811 AC	CLEAR AC	
24/07/2018	ROUTINE	0.020 AC		27 QU	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.022 AC		45 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.022 AC		36 AC	1.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.022 AC		36 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.022 AC		30 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.022 AC		32 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.022 AC		32 AC	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.022 AC	36.8 AC	37 AC	1.5 AC	0.8889 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	18	21157	64224	4	129	1676	ND	1	ND	152	87209
06/12/2017	20	15791	63530	5	275	2783	ND	4	ND	304	82408
28/06/2018	32	22904	67139	4	249	3112	ND	5	ND	290	93445
20/06/2019	39	11205	66397	21	439	5998	11	9	ND	519	84119

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
22/05/2018	33	32	2.62	D
28/06/2018	28	16	1.89	B
24/07/2018	33	30	2.46	C
25/08/2018	28	16	1.89	B
25/09/2018	30	19	2.00	B
23/10/2018	37	23	1.50	B
22/11/2018	36	25	1.77	B
08/12/2018	37	24	1.56	B
21/05/2019	35	23	1.63	B
20/06/2019	34	24	1.97	B

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-2**LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	3	ND	ND	3
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	0.01	0.01
28/06/2018	ND	ND	0.02
20/06/2019	0.01	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.350 AC
06/12/2017	0.000 AC	0.270 AC
28/06/2018	0.000 AC	0.490 AC
20/06/2019	0.000 AC	0.680 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC
06/12/2017	0.24 AC
28/06/2018	0.20 AC
20/06/2019	0.20 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1A	LIGHT ORANGE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
14.	TRAFO 1 ESP UNIT 2	10037	17

A) Analysis

- **Acidity** of 0.031mgKOH/gm is **ACCEPTABLE**, while **IFT** of 31.0mN/m is **QUESTIONABLE**. Low interfacial tension value is an indication that there are contaminants in the mineral oil that is weakening the molecular bonds of the mineral oil.
- **Dielectric** of 62kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 13ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.37% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- Retest **Acidity** and **IFT** in 12 months to determine if transformer oil deteriorate further to a level that require oil corrective work. **Liquid Power Factor** could also be tested to assess the effect of the contaminants detected on the dielectric properties of the transformer oil.
- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.025 AC		24 UN	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.025 AC	32.9 AC	77 AC	1.0 AC	0.8693 AC	CLEAR AC	
24/07/2018	ROUTINE	0.028 AC		69 AC	1.0 AC		CLEAR AC	
25/08/2018	ROUTINE	0.031 AC		74 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.028 AC		74 AC	1.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.028 AC		54 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.028 AC		54 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.028 AC		46 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.031 AC		67 AC	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.031 AC	31.0 QU	62 AC	1.0 AC	0.8776 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	93	17225	43231	5	57	775	ND	2	ND	157	61388
20/06/2019	62	18560	46166	5	50	839	ND	ND	ND	117	65682

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	32	3.04	E
28/06/2018	28	11	1.30	B
24/07/2018	31	11	1.04	B
25/08/2018	29	12	1.26	B
25/09/2018	30	10	1.05	B
23/10/2018	31	11	1.04	B
22/11/2018	32	10	0.95	B
08/12/2018	30	10	1.05	B
21/05/2019	30	12	1.26	B
20/06/2019	30	13	1.37	C

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	0.06	0.03
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.030 AC	0.420 AC
20/06/2019	0.000 AC	0.720 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	2C	LAVENDER BLUE
20/06/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
15.	TRAFO 2 ESP UNIT 2	10036	18

A) Analysis

- **Acidity** of 0.065mgKOH/g and **IFT** of 29.7mN/m are **QUESTIONABLE**.
Although the level remained stable, the oil are containing oxidation by-products. These oxidation by-products are depositing and degrading the oil as well as the paper insulation. Therefore, removal of these contaminations are highly recommended.
- **Dielectric** of 76kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 12ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.14% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in magenta, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.053 QU		55 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.056 QU	31.3 QU	66 AC	1.0 AC	0.8573 AC	CLEAR AC	
24/07/2018	ROUTINE	0.059 QU		79 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.062 QU		67 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.056 QU		51 AC	1.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.062 QU		67 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.062 QU		61 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.062 QU		59 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.065 QU		77 AC	1.0 AC		CLEAR AC	
20/06/2019	ROUTINE	0.065 QU	29.7 QU	76 AC	1.0 AC	0.8655 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	8	22714	64906	5	67	879	ND	ND	ND	80	88579
20/06/2019	5	29439	65338	5	25	977	ND	ND	ND	35	95789

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	16	1.52	C
28/06/2018	27	13	1.53	C
24/07/2018	31	8	0.76	A
25/08/2018	31	10	0.95	B
25/09/2018	29	9	0.94	B
23/10/2018	31	13	1.24	B
22/11/2018	33	8	0.66	A
08/12/2018	30	8	0.84	A
21/05/2019	30	11	1.15	B
20/06/2019	31	12	1.14	B

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	42	ND	ND	ND	42
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.680 AC
20/06/2019	0.000 AC	0.450 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
16.	TRAFO 3 ESP UNIT 2	10033	19

A) Analysis

- **Acidity** of 0.073mgKOH/g is **QUESTIONABLE** while **IFT** of 26.5mN/m is **UNACCEPTABLE**. Although the level remained stable, the oil are containing oxidation by-products. These oxidation by-products are depositing and degrading the oil as well as the paper insulation. Therefore, removal of these contaminations are highly recommended.
- **Dielectric** of 53kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 13ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.37% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in magenta, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
21/05/2018	OFFLINE			NORMAL		GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.059 QU		51 AC	1.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.062 QU	26.3 UN	78 AC	1.5 AC	0.8602 AC	CLEAR AC	
24/07/2018	ROUTINE	0.070 QU		67 AC	1.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.067 QU		73 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.067 QU		59 AC	1.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.070 QU		86 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.070 QU		13 UN	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.070 QU		8 UN	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.073 QU		50 AC	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.073 QU	26.5 UN	53 AC	1.5 AC	0.8687 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	14	17627	48672	4	47	935	ND	ND	ND	65	67299
20/06/2019	16	17449	51923	5	58	1049	ND	ND	ND	79	70500

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	18	1.71	D
28/06/2018	28	14	1.65	C
24/07/2018	32	12	1.14	B
25/08/2018	30	13	1.37	C
25/09/2018	30	12	1.26	B
23/10/2018	32	13	1.24	B
22/11/2018	33	84	6.89	E
08/12/2018	31	84	7.98	E
21/05/2019	30	13	1.37	C
20/06/2019	30	13	1.37	C

FIELD SERVICE

DATE	SERVICE
------	---------

DATE PRINTED 19/07/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.570 AC
20/06/2019	0.010 AC	0.790 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN
20/06/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
17.	TRAFO 4 ESP UNIT 2	10032	20

A) Analysis

- **Acidity** of 0.036mgKOH/gm is **ACCEPTABLE**, while **IFT** of 30.3mN/m is **QUESTIONABLE** Low interfacial tension value is an indication that there are contaminants in the mineral oil that is weakening the molecular bonds of the mineral oil.
- **Dielectric** of 61kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 14ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.47% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.01% by weight is detected, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in lavender blue, thus **Non-corrosive of 2C**.

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B) Recommendation

- Retest **Acidity** and **IFT** in 12 months to determine if transformer oil deteriorate further to a level that require oil corrective work. **Liquid Power Factor** could also be tested to assess the effect of the contaminants detected on the dielectric properties of the transformer oil.
- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
06/12/2017	OFFLINE						
21/05/2018	OFFLINE					GOOD	NONE
28/06/2018	ONLINE						
24/07/2018	OFFLINE						
25/08/2018	OFFLINE						
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2018	ROUTINE	0.025 AC		40 AC	1.0 AC		CLEAR AC	
28/06/2018	ROUTINE	0.028 AC	32.6 AC	88 AC	1.0 AC	0.8648 AC	CLEAR AC	
24/07/2018	ROUTINE	0.031 AC		59 AC	1.0 AC		CLEAR AC	
25/08/2018	ROUTINE	0.031 AC		68 AC	1.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.031 AC		51 AC	1.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.034 AC		57 AC	1.0 AC		CLEAR AC	
22/11/2018	ROUTINE	0.034 AC		69 AC	1.0 AC		CLEAR AC	
08/12/2018	ROUTINE	0.034 AC		61 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.036 AC		60 AC	1.0 AC		CLEAR AC	
20/06/2019	ROUTINE	0.036 AC	30.3 QU	61 AC	1.0 AC	0.8732 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	22	25201	74464	5	85	823	ND	ND	ND	112	100600
20/06/2019	33	25208	70145	4	35	853	ND	ND	ND	72	96278

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2018	31	18	1.71	D
28/06/2018	27	15	1.77	D
24/07/2018	32	13	1.24	B
25/08/2018	31	10	0.95	B
25/09/2018	29	11	1.15	B
23/10/2018	32	15	1.42	C
22/11/2018	33	14	1.15	B
08/12/2018	31	18	1.71	D
21/05/2019	31	14	1.33	B
20/06/2019	30	14	1.47	C

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	7	ND	ND	ND	7

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.03
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	1.620 AC
20/06/2019	0.020 AC	1.150 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	0.01 UN
20/06/2019	0.01 UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	2C	LAVENDER BLUE
20/06/2019	NON-COR	2C	LAVENDER BLUE

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
18.	MAIN TRAFO UNIT 2	1006902	21

A) Analysis

- **Acidity** of 0.011mgKOH/g and **IFT** of 39.3mN/m are **ACCEPTABLE**.
- **Dielectric** of 58kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 11ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.78% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.22% by weight is **ACCEPTABLE**.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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DATE PRINTED 19/07/2019
ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAFU UNIT 2

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
06/12/2017	ONLINE						
22/05/2018	ONLINE			NORMAL		GOOD	NONE
28/06/2018	OFFLINE	50	45				
24/07/2018	ONLINE					GOOD	
25/08/2018	ONLINE						
25/09/2018	ONLINE						
23/10/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 170000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
22/05/2018	ROUTINE	0.011 AC		46 AC	0.5 AC		CLEAR AC	
28/06/2018	ROUTINE	0.008 AC	42.1 AC	75 AC	0.5 AC	0.8663 AC	CLEAR AC	
24/07/2018	ROUTINE	0.011 AC		79 AC	0.5 AC		CLEAR AC	
25/08/2018	ROUTINE	0.011 AC		83 AC	0.5 AC		CLEAR AC	
25/09/2018	ROUTINE	0.011 AC		90 AC	0.5 AC		CLEAR AC	
23/10/2018	ROUTINE	0.011 AC		64 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.011 AC		61 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.011 AC		57 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.011 AC		63 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.011 AC	39.3 AC	58 AC	0.5 AC	0.8745 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	19	12733	53386	10	193	2448	ND	2	ND	224	68791
20/06/2019	10	17382	50681	8	107	712	ND	2	ND	127	68902

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
22/05/2018	33	14	1.15	B
28/06/2018	28	10	1.18	B
24/07/2018	33	11	0.90	B
25/08/2018	34	9	0.74	A
25/09/2018	32	7	0.67	A
23/10/2018	29	7	0.74	A
22/11/2018	32	7	0.67	A
08/12/2018	30	6	0.63	A
21/05/2019	35	7	0.50	A
20/06/2019	35	11	0.78	A

FIELD SERVICE

DATE	SERVICE
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DATE PRINTED 19/07/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - MAIN TRAF0 UNIT 2

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 170000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	113	113
20/06/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
28/06/2018	ND	ND	0.03
20/06/2019	ND	ND	0.02

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.410 AC
20/06/2019	0.010 AC	0.730 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	0.22 AC
20/06/2019	0.22 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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3.0 Summary of Our Comments & Recommendations:

No.	Sub Name	Unit No.	TC	Recommendation
1.	MAIN TRAFO UNIT 1	100691	4	- Add passivator - Periodic condition monitoring
2.	TRAFO 1 ESP UNIT 1	10937	5	- Retest 6 months: Dielectric and Water Content - Add inhibitor 2,6-DBPC and passivator - Other parameters: Periodic condition monitoring
3.	TRAFO 2 ESP UNIT 1	10035	6	Perform OIL REGENERATION
4.	TRAFO 3 ESP UNIT 1	10031	7	Perform OIL REGENERATION
5.	TRAFO 4 ESP UNIT 1	10034	8	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
6.	MAIN TRAFO UNIT 3	3011160192	23	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
7.	UAT 3	PX18-0082	22	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
8.	TRAFO 1 ESP UNIT 3	0073084208	11	- Retest 12 months: Specific Gravity - Add passivator - Other parameters: Periodic condition monitoring
9.	TRAFO 2 ESP UNIT 3	0070084208	12	- Retest 12 months: Specific Gravity - Add passivator - Other parameters: Periodic condition monitoring

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No.	Sub Name	Unit No.	TC	Recommendation
10.	TRAFO 3 ESP UNIT 3	0072084208	13	- Retest 12 months: Specific Gravity - Retest 6 months: Dielectric and Water Content - Add passivator - Other parameters: Periodic condition monitoring
11.	TRAFO 4 UNIT 3	0071084208	14	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
12.	ASH HANDLING-1	10820108	15	- Retest 6 months: Dielectric and Water Content - Add passivator - Other parameters: Periodic condition monitoring
13.	ASH HANDLING-2	10820107	16	- Add passivator - Periodic condition monitoring
14.	TRAFO 1 ESP UNIT 2	10037	17	- Retest 12 months: Acidity, IFT, and Liquid Power Factor - Add inhibitor 2,6-DBPC and passivator - Other parameters: Periodic condition monitoring
15.	TRAFO 2 ESP UNIT 2	10036	18	Perform OIL REGENERATION
16.	TRAFO 3 ESP UNIT 2	10033	19	Perform OIL REGENERATION

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No.	Sub Name	Unit No.	TC	Recommendation
17.	TRAFO 4 ESP UNIT 2	10032	20	• Retest 12 months: Acidity, IFT, and Liquid Power Factor • Add inhibitor 2,6-DBPC and passivator • Other parameters: Periodic condition monitoring
18.	MAIN TRAFO UNIT 2	1006902	21	• Add passivator • Periodic condition monitoring

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no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium															
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612						
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida		
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	
1	1	27/02/17	Main trafo	1	1006901	2011	30.857	Wolong	0,014	39,9	99	0,5	0,8693	C		9	A	135	15.834	57.625	27	807	3.099
	4	04/07/17							0,017	41,5	58	0,5	0,8691	C		10	A	147	9.471	4.328	32	655	2.728
	4	30/07/2017							0,017		70	0,5		C		8	B						
	4	30/08/2017							0,017		92	0,5		C		6	A						
	4	29/09/2017							0,014		82	0,5		C		8	A						
	4	30/10/2017							0,014		82	0,5		C		6	A						
	4	27/11/2017							0,014		53	0,5		C		8	A						
	4	07/12/2017							0,011	40,1	87	0,5	0,8745	C		7	A	183	12.895	43.794	39	715	3.166
	4	21/05/2018							0,011		77	0,5		C		10	B						
	4	28/06/2018							0,014	41,9	69	0,5	0,8716	C		7	A	176	11.044	46.776	41	758	3.393
	4	24/07/2018							0,014		86	0,5		C		6	A						
	4	25/08/2018							0,017		90	0,5		C		6	A						
	4	25/09/2018							0,014		96	0,5		C		7	A						
	4	23/10/2018							0,014		83	0,5		C		8	A						
	4	22/11/2018							0,014		73	0,5		C		9	A						
	4	08/12/2018							0,014		75	0,5		C		8	A						
	4	21/05/2019							0,014		71	0,5		C		8	A						
	4	20/06/2019							0,017	41,3	66		0,8789	C		7	A		19.122	60.738	26	756	3.272
2	5	04/07/17	Trafo 1 ESP	1	10937	2010	1.143	Fujia	0,056	25,3	50	1,5	0,8529	C		18	D	35	20.494	67.487	4	69	1.143
	5	30/07/2017							0,059		57	1,5		C		20	D						
	5	30/08/2017							0,056		65	1,5		C		18	D						
	5	29/09/2017							0,059		58	1,5		C		19	D						
	5	30/10/2017							0,059		23	1,5		C		32	C						
	5	27/11/2017							0,062		25	1,5		C		33	C						
	5	07/12/2017							0,062	25,0	49	1,5	0,8593	C		18	D	39	19.933	67.198	4	83	1.261
	5	21/05/2018							0,065		38	1,5		C		33	E						
	5	28/06/2018							0,067	25,9	32	1,5	0,8560	C		30	E	44	20.039	64.529	4	119	1.696
	5	24/07/2018							0,073		38	1,5		C		19	D						
	5	25/08/2018							0,020		71	0,5		C		12	B						
	5	25/09/2018							0,017		63	0,5		C		18	B						
	5	23/10/2018							0,020		49	0,5		C		24	B						
	5	22/11/2018							0,017		30	0,5		C		23	C						
	5	08/12/2018							0,017		41	0,5		C		18	D						
	5	21/05/2019							0,017		40	0,5		C		22	C						
	5	20/06/2019							0,017	38,7	34	0,5	0,8747	C		23	C	22	27.043	68.613	5		689
3	6	04/07/17	Trafo 2 ESP	1	10035	2010	1.143	Fujia	0,042	26,6	41	1,5	0,8540	C		21	D	27	18.639	66.781	3	64	1.106
	6	30/07/2017							0,045		25	1,5		C		27	E						
	6	30/08/2017							0,042		42	1,5		C		14	C						
	6	29/09/2017							0,048		59	1,5		C		19	D						
	6	30/10/2017							0,051		62	1,5		C		17	B						
	6	27/11/2017							0,051		50	1,5		C		18	B						
	6	07/12/2017							0,063	27,3	51	1,5	0,8589	C		17	D	20	19.360	61.590	3	45	729
	6	21/05/2018							0,056		45	1,5		C		20	D						
	6	28/06/2018							0,059	27,6	55	1,5	0,8570	C		17	C	22	22.494	64.443	4	57	789
	6	24/07/2018							0,062		47	1,5		C		15	C						
	6	25/08/2018							0,022		56	0,5		C		13	B						
	6	25/09/2018							0,020		67	0,5		C		11	B						
	6	23/10/2018							0,020		71	0,5		C		14	B						
	6	22/11/2018							0,020		63	0,5		C		18	C						



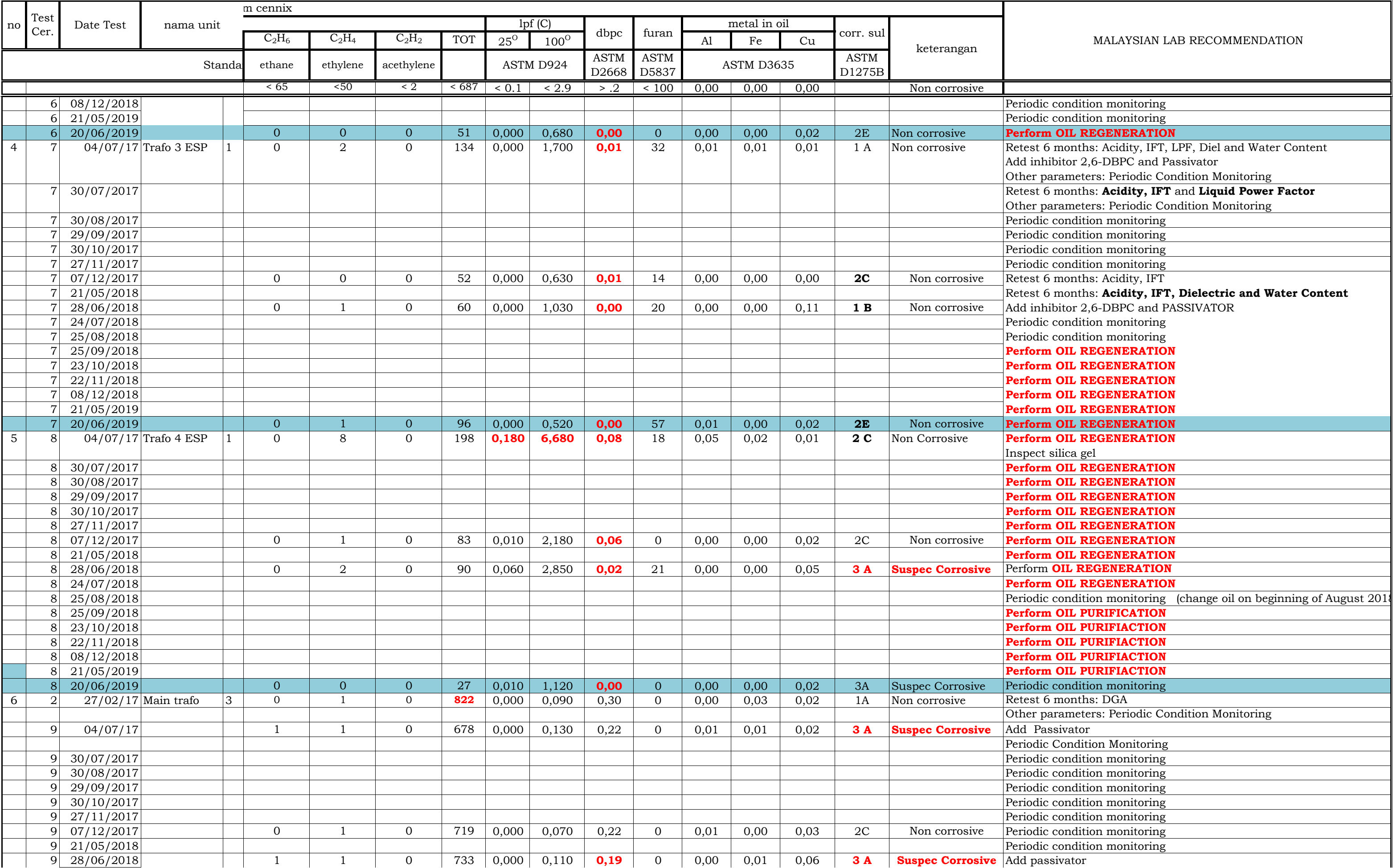
no	Test Cer.	Date Test	nama unit	m cennix											corr. sul	keterangan	MALAYSIAN LAB RECOMMENDATION	
								lpf (C)		dbpc	furan	metal in oil						
				C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu				
Standar				ethane	ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B			
					< 65	<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0,00	0,00	0,00		Non corrosive	
1	1	27/02/17	Main trafo	1	3	4	0	976	0,000	0,070	0,32	0	0,00	0,00	0,01	1B	Non Corrosive	Retest 6 months: DGA Check leakages Other parameters: Periodic Condition Monitoring
	4	04/07/17			4	5	0	843	0,000	0,140	0,24	0	0,02	0,01	0,01	3 A	Suspec Corrosive	Retest 6 months: DGA Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring
	4	30/07/2017																Periodic condition monitoring
	4	30/08/2017																Periodic condition monitoring
	4	29/09/2017																Periodic condition monitoring
	4	30/10/2017																Periodic condition monitoring
	4	27/11/2017																Periodic condition monitoring
	4	07/12/2017			5	5	0	947	0,000	0,040	0,27	0	0,00	0,01	0,04	2C	Non corrosive	Retest 6 mnth : DGA
	4	21/05/2018																Periodic condition monitoring
	4	28/06/2018			5	6	0	986	0,000	0,140	0,24	0	0,00	0,00	0,04	3 A	Suspec Corrosive	Add passivator
	4	24/07/2018																Periodic condition monitoring
	4	25/08/2018																Periodic condition monitoring
	4	25/09/2018																Periodic condition monitoring
	4	23/10/2018																Periodic condition monitoring
	4	22/11/2018																Periodic condition monitoring
	4	08/12/2018																Periodic condition monitoring
	4	21/05/2019																Periodic condition monitoring
	4	20/06/2019			2	3	0	805	0,000	0,180	0,24	0	0,01	0,00	0,02	3A	Suspec Corrosive	Add passivator
2	5	04/07/17	Trafo 1 ESP	1	0	1	0	109	0,000	1,870	0,00	95	0,00	0,01	0,01	1 B	Non Corrosive	Perform OIL REGENERATION
	5	30/07/2017																Perform OIL REGENERATION
	5	30/08/2017																Perform OIL REGENERATION
	5	29/09/2017																Perform OIL REGENERATION
	5	30/10/2017																Perform OIL REGENERATION
	5	27/11/2017																Perform OIL REGENERATION
	5	07/12/2017			0	1	0	127	0,000	0,570	0,00	75	0,00	0,00	0,00	1A	Non corrosive	Perform OIL REGENERATION
	5	21/05/2018																Perform OIL REGENERATION
	5	28/06/2018			0	2	0	169	0,000	0,680	0,00	85	0,00	0,00	0,03	1 B	Non corrosive	Perform OIL REGENERATION
	5	24/07/2018																Perform OIL REGENERATION
	5	25/08/2018																Periodic condition monitoring (change oil on beginning of August 2018)
	5	25/09/2018																Periodic condition monitoring
	5	23/10/2018																Periodic condition monitoring
	5	22/11/2018																Preparing for Purification
	5	08/12/2018																Preparing for Purification
	5	21/05/2019																Preparing for Purification
	5	20/06/2019			0	0	0	63	0,000	0,230	0,00	0	0,00	0,00	0,02	1B	Non corrosive	Preparing for Purification
3	6	04/07/17	Trafo 2 ESP	1	0	1	0	95	0,000	1,900	0,02	62	0,00	0,01	0,00	1 A	Non Corrosive	Perform OIL REGENERATION
	6	30/07/2017																Perform OIL REGENERATION
	6	30/08/2017																Perform OIL REGENERATION
	6	29/09/2017																Perform OIL REGENERATION
	6	30/10/2017																Perform OIL REGENERATION
	6	27/11/2017																Perform OIL REGENERATION
	6	07/12/2017			0	1	0	69	0,000	0,980	0,02	67	0,00	0,00	0,02	1B	Non corrosive	Perform OIL REGENERATION
	6	21/05/2018																Perform OIL REGENERATION
	6	28/06/2018			0	0	0	83	0,000	1,130	0,00	84	0,00	0,00	0,02	1 B	Non corrosive	Perform OIL REGENERATION
	6	24/07/2018																Perform OIL REGENERATION
	6	25/08/2018																Periodic condition monitoring (change oil on beginning of August 2018)
	6	25/09/2018																Periodic condition monitoring
	6	23/10/2018																Periodic condition monitoring
	6	22/11/2018																Periodic condition monitoring



no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION	
Standar					
1	1	27/02/17	Main trafo	1	Fr Test 20 Juni 2019 Ada penurunan kondisi trafo ESP terutama Unit 1 dan 2 Jadi kami sarankan untuk ESP 1 Unit 1 dan Ash Handling 1 segera dilakukan PURIFIKASI dengan temperatur sekitar 700 GANTI MINYAK pada trafo ESP 3 Unit 1 dan ESP 2 dan 3 Unit 2 Monitoring kualitas minyak untuk ESP 1 dan 4 Unit 2 For TEST 21 Mei 2019 Ada beberapa perubahan kondisi trafo seperti, Main Trafo dan UAT Unit 3 yang ganti Trafo baru serta perbaikan oli diESP 3 Unit 2 Jadi kami sarankan untuk ESP 1 dan 4 Unit 1, ESP 2 Unit 3 dan Ash Handling 1 segera dilakukan PURIFIKASI dengan temperatur sekitar 700 GANTI MINYAK pada trafo ESP 3 Unit 1 dan ESP 2 dan 3 Unit 2 For TEST 08 Desember 2018 Pada dasarnya kondisi masih sama dengan pada saat pengetesan bulan November 2018, <u>Untuk trafo ESP 3 Unit 2 kondisi BDV sudah sangat rendah sekali, harap menjadikan perhatian khusus</u> Kami sarankan untuk Untuk ESP 1 dan 4 Unit 1, ESP 2 Unit 3 dan Ash Handling 1 segera dilakukan PURIFIKASI dengan temperatur sekitar 70 ⁰ GANTI MINYAK pada trafo ESP 3 Unit 1 dan ESP 2 Unit 2 For TEST 22 November 2018 Meskipun hampir semua trafo dalam keadaan prima, beberapa sudah mulai mengalami penurunan kondisi. PURIFIKASI Untuk ESP 1 dan 4 Unit 1, ESP 2 Unit 3 dan Ash Handling 1 ditemukan telah mengalami trend penurunan fungsi BDV dan Water Content. Purifikasi sebaiknya dilakukan dengan temperatur sekitar 70⁰ agar mampu menghilangkan kandungan air dalam isolasi kertas. GANTI MINYAK Pada trafo ESP 3 Unit 1, ESP 2 dan 3 Unit 2 perlu dilakukan GANTI MINYAK, mengingat tingkat Acidity nya yang sudah tinggi.
	4	04/07/17			
	4	30/07/2017			
	4	30/08/2017			
	4	29/09/2017			
	4	30/10/2017			
	4	27/11/2017			
	4	07/12/2017			
	4	21/05/2018			
	4	28/06/2018			
	4	24/07/2018			
	4	25/08/2018			
	4	25/09/2018			
	4	23/10/2018			
	4	22/11/2018			
	4	08/12/2018			
	4	21/05/2019			
	4	20/06/2019			
2	5	04/07/17	Trafo 1 ESP	1	
	5	30/07/2017			
	5	30/08/2017			
	5	29/09/2017			
	5	30/10/2017			
	5	27/11/2017			
	5	07/12/2017			
	5	21/05/2018			
	5	28/06/2018			
	5	24/07/2018			
	5	25/08/2018			
	5	25/09/2018			
	5	23/10/2018			
	5	22/11/2018			
	5	08/12/2018			
	5	21/05/2019			
	5	20/06/2019			
3	6	04/07/17	Trafo 2 ESP	1	
	6	30/07/2017			
	6	30/08/2017			
	6	29/09/2017			
	6	30/10/2017			
	6	27/11/2017			
	6	07/12/2017			
	6	21/05/2018			
	6	28/06/2018			
	6	24/07/2018			
	6	25/08/2018			
	6	25/09/2018			
	6	23/10/2018			
	6	22/11/2018			



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium															
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612						
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida		
		BATASAN OPERASI						< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	
	6	08/12/2018						0,020		51	0,5		C		14	B							
	6	21/05/2019							0,020		71	0,5		C		15	B						
	6	20/06/2019						0,020	27,9	68	0,5	0,8658	C		17	B	23	17.710	61.707	5	23	842	
4	7	04/07/17	Trafo 3 ESP	1	10031	2010	1.143	Fujia	0,048	28,4	44	1,0	0,8537	C		32	E	32	15.273	65.111	5	95	1.002
	7	30/07/2017							0,042		77	1,5			10	D							
	7	30/08/2017							0,042		61	1,5		C		13	C						
	7	29/09/2017							0,045		72	1,5		C		14	B						
	7	30/10/2017							0,048		72	1,5		C		16	B						
	7	27/11/2017							0,051		62	1,5		C		17	C						
	7	07/12/2017							0,063	29,3	57	1,5	0,8624	C		16	C	7	14.730	65.738	3	42	617
	7	21/05/2018							0,056		48	1,5		C		21	D						
	7	28/06/2018							0,053	29,7	62	1,5	0,8605	C		17	D	9	17.958	66.007	4	46	781
	7	24/07/2018							0,510		64	1,5		C		17	D						
	7	25/08/2018							0,056		57	1,5		C		15	C						
	7	25/09/2018							0,051		46	1,5		C		21	D						
	7	23/10/2018							0,056		44	1,5		C		20	C						
	7	22/11/2018							0,056		39	1,5		C		22	C						
	7	08/12/2018							0,056		39	1,5		C		20	C						
	7	21/05/2019							0,051		33	1,5		C		19	B						
	7	20/06/2019							0,051	29,6	30	1,5	0,8689	C		19	B	16	17.771	65.478	6	73	829
5	8	04/07/17	Trafo 4 ESP	1	10034	2010	1.143	Fujia	0,028	26,3	24	1,5	0,8596	C		29	E	13	12.265	66.770	5	172	1.206
	8	30/07/2017							0,022		88	1,5		C		15	D						
	8	30/08/2017							0,028		44	1,5		C		13	C						
	8	29/09/2017							0,031		53	1,0		C		15	B						
	8	30/10/2017							0,028		48	1,0		C		18	B						
	8	27/11/2017							0,034		41	1,5		C		30	D						
	8	07/12/2017							0,031	27,2	41	1,5	0,8645	C		18	D	9	13.331	63.527	3	70	720
	8	21/05/2018							0,034		37	1,5		C		23	D						
	8	28/06/2018							0,037	27,5	55	1,5	0,8653	C		20	D	9	17.962	64.248	4	75	1.054
	8	24/07/2018							0,039		50	1,5		C		19	D						
	8	25/08/2018							0,017		61	0,5		C		14	C						
	8	25/09/2018							0,014		45	0,5		C		20	D						
	8	23/10/2018							0,014		52	0,5		C		21	B						
	8	22/11/2018							0,014		44	0,5		C		21	C						
	8	08/12/2018							0,014		47	0,5		C		20	B						
	8	21/05/2019							0,014		47	0,5		C		19	B						
	8	20/06/2019							0,014	38,6	45	0,5	0,8775	C		19	B	11	15.056	64.828	5	11	951
6	2	27/02/17	Main trafo	3	1122170	2010	26.706	Shandong	0,017	39,7	87	0,5	0,8781	C		9	A	4	13.330	43.140	9	808	2.878
	9	04/07/17							0,017	42,3	78	0,5	0,8780	C		15	B	24	10.131	44.084	11	641	2.250
	9	30/07/2017							0,014		77	0,5		C		17	B						
	9	30/08/2017							0,017		86	0,5		C		8	A						
	9	29/09/2017							0,014		86	0,5		C		10	A						
	9	30/10/2017							0,014		88	0,5		C		9	A						
	9	27/11/2017							0,014		87	0,5		C		8	A						
	9	07/12/2017							0,014	41,5	84	0,5	0,8932	C			A	39	14.023	43.226	11	668	2.456
	9	21/05/2018							0,014		58	0,5		C		11	B						
	9	28/06/2018							0,014	41,0	85	0,5	0,8805	C		8	A	18	11.431	48.283	13	700	1.512

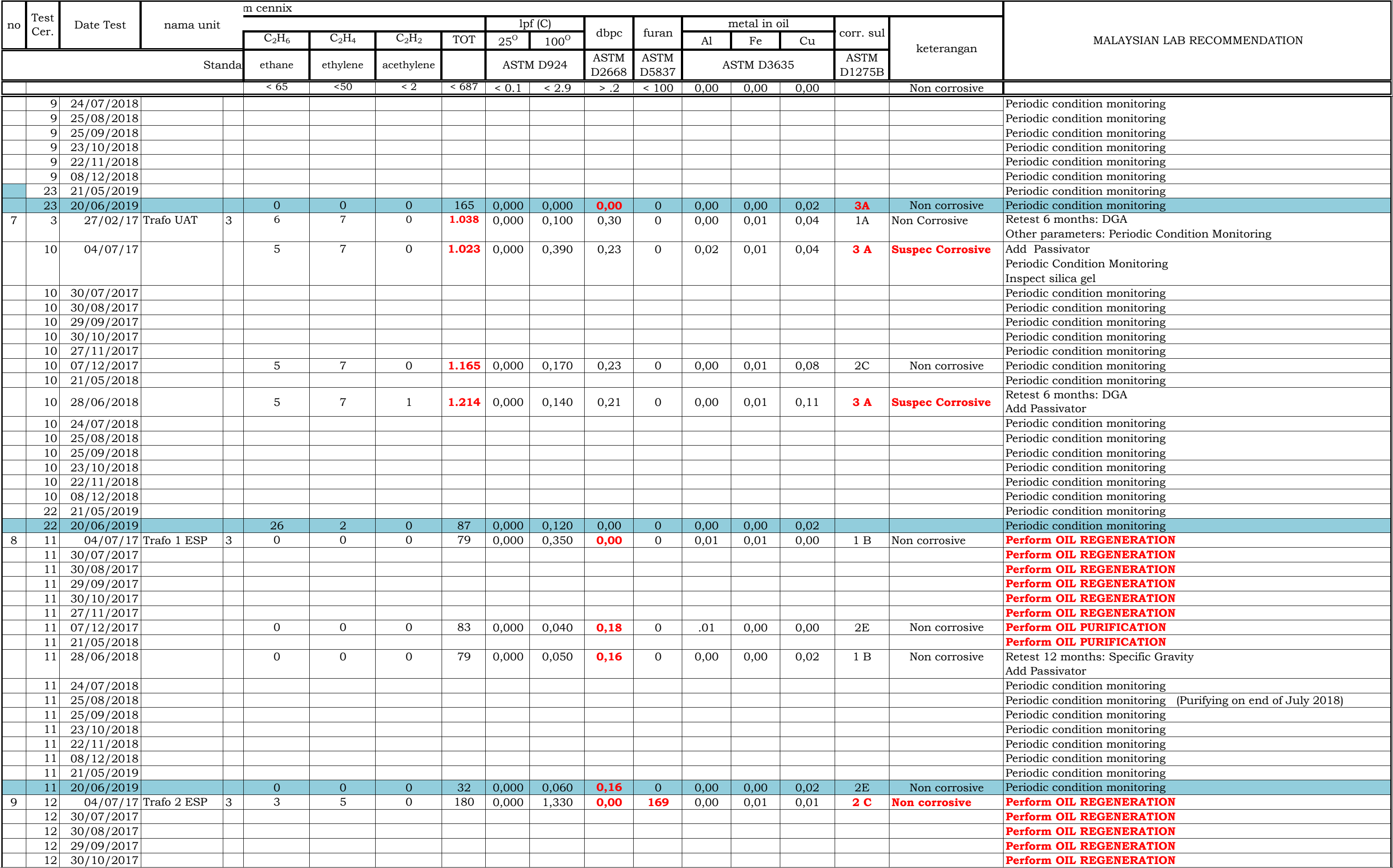




no	Test Cer.	Date Test	nama unit		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar					
	6	08/12/2018			
	6	21/05/2019			
	6	20/06/2019			
4	7	04/07/17	Trafo 3 ESP	1	
	7	30/07/2017			
	7	30/08/2017			
	7	29/09/2017			
	7	30/10/2017			
	7	27/11/2017			
	7	07/12/2017			
	7	21/05/2018			
	7	28/06/2018			
	7	24/07/2018			
	7	25/08/2018			
	7	25/09/2018			
	7	23/10/2018			
	7	22/11/2018			
	7	08/12/2018			
	7	21/05/2019			
	7	20/06/2019			
5	8	04/07/17	Trafo 4 ESP	1	
	8	30/07/2017			
	8	30/08/2017			
	8	29/09/2017			
	8	30/10/2017			
	8	27/11/2017			
	8	07/12/2017			
	8	21/05/2018			
	8	28/06/2018			
	8	24/07/2018			
	8	25/08/2018			
	8	25/09/2018			
	8	23/10/2018			
	8	22/11/2018			
	8	08/12/2018			
	8	21/05/2019			
	8	20/06/2019			
6	2	27/02/17	Main trafo	3	
	9	04/07/17			
	9	30/07/2017			
	9	30/08/2017			
	9	29/09/2017			
	9	30/10/2017			
	9	27/11/2017			
	9	07/12/2017			
	9	21/05/2018			
	9	28/06/2018			



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium															
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612						
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida		
		BATASAN OPERASI							< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500
	9	24/07/2018						0,014		77	0,5		C		8	A							
	9	25/08/2018						0,017		82	0,5		C		7	A							
	9	25/09/2018						0,017		83	0,5		C		6	A							
	9	23/10/2018						0,017		78	0,5		C		8	A							
	9	22/11/2018						0,014		63	0,5		C		7	A							
	9	08/12/2018						0,014		83	0,5		C		7	A							
	23	21/05/2019		3011160192	2018	30.183	PAUWES TRAF	0,011		93	0,5		C		7	A							
	23	20/06/2019	Trafo UAT	3	1035082	2010	4.749	Shandong	0,011	40,6	83	0,5	0,8786	C		8	A	3	9.903	31.346	6	156	369
7	3	27/02/17	Trafo UAT	3	1035082	2010	4.749	Shandong	0,017	37,6	94	0,5	0,8768	C		9	A	20	12.535	56.879	20	985	3.877
	10	04/07/17						0,022	38,1	60	0,5	0,8767	C		11	A	20	16.028	68.308	20	971	3.536	
	10	30/07/2017						0,025		88	0,5		C		14	B							
	10	30/08/2017						0,022		89	0,5		C		11	A							
	10	29/09/2017						0,020		80	0,5		C		12	A							
	10	30/10/2017						0,022		86	0,5		C		11	A							
	10	27/11/2017						0,022		86	0,5		C		12	A							
	10	07/12/2017						0,020	37,6	86	0,5	0,8200	C		12	A	36	18.752	66.758	1	1.116	3.622	
	10	21/05/2018						0,020		81	0,5		C		14	B							
	10	28/06/2018						0,022	37,5	79	0,5	0,8793	C		10	A	35	13.798	68.180	22	1.144	3.644	
	10	24/07/2018						0,020		76	0,5		C		11	A							
	10	25/08/2018						0,022		78	0,5		C		8	A							
	10	25/09/2018						0,020		75	0,5		C		8	A							
	10	23/10/2018						0,020		69	0,5		C		8	A							
	10	22/11/2018						0,022		80	0,5		C		11	A							
	10	08/12/2018						0,020		61	0,5		C		8	A							
	22	21/05/2019		PX18-0082	2018	12.000	B & D	0,014		85	0,5		C		5	A							
	22	20/06/2019	Trafo 1 ESP	3	007308428	2010	731	Zhejiang	0,014	37,0	89	0,5	0,8798	C		5	A	4	10.251	34.223	16	39	232
8	11	04/07/17	Trafo 1 ESP	3	007308428	2010	731	Zhejiang	0,022	23,2	32	1,5	0,8432	C		26	D	14	12.943	68.807	3	62	1.243
	11	30/07/2017						0,028		47	1,5		C		21	D							
	11	30/08/2017						0,028		36	1,5		C		22	D							
	11	29/09/2017						0,025		45	1,5		C		20	C							
	11	30/10/2017						0,025		40	1,5		C		19	D							
	11	27/11/2017						0,025		31	1,5		C		34	C							
	11	07/12/2017						0,022	34,6	31	1,5	0,8190	C		31	C	14	12.739	65.097	3	66	612	
	11	21/05/2018						0,022		31	0,5		C		27	D							
	11	28/06/2018						0,022	35,0	60	0,5	0,8154	C		10	B	19	15.552	64.448	3	57	1.391	
	11	24/07/2018						0,020		53	0,5		C		9	B							
	11	25/08/2018						0,022		77	0,5		C		9	B							
	11	25/09/2018						0,020		64	0,5		C		10	B							
	11	23/10/2018						0,020		63	0,5		C		12	B							
	11	22/11/2018						0,020		53	0,5		C		9	B							
	11	08/12/2018						0,020		62	0,5		C		9	B							
	11	21/05/2019						0,022		85	0,5		C		9	B							
	11	20/06/2019	Trafo 2 ESP	3	007008428	2010	731	Zhejiang	0,022	35,2	65	0,5	0,8212	C		9	B	11	19.448	64.974	6	15	1.064
9	12	04/07/17	Trafo 2 ESP	3	007008428	2010	731	Zhejiang	0,118	16,2	24	3,0	0,8435	C		39	D	38	16.781	70.523	5	129	2.431
	12	30/07/2017						0,112		28	3,0		C		38	E							
	12	30/08/2017						0,109		34	2,5		C		17	D							
	12	29/09/2017						0,115		21	2,5		C		37	D							
	12	30/10/2017						0,118		33	2,5		C		20	D							





no	Test Cer.	Date Test	nama unit		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar					
	9	24/07/2018			
	9	25/08/2018			
	9	25/09/2018			
	9	23/10/2018			
	9	22/11/2018			
	9	08/12/2018			
	23	21/05/2019			
	23	20/06/2019			
7	3	27/02/17	Trafo UAT	3	
	10	04/07/17			
	10	30/07/2017			
	10	30/08/2017			
	10	29/09/2017			
	10	30/10/2017			
	10	27/11/2017			
	10	07/12/2017			
	10	21/05/2018			
	10	28/06/2018			
	10	24/07/2018			
	10	25/08/2018			
	10	25/09/2018			
	10	23/10/2018			
	10	22/11/2018			
	10	08/12/2018			
	22	21/05/2019			
	22	20/06/2019			
8	11	04/07/17	Trafo 1 ESP	3	
	11	30/07/2017			
	11	30/08/2017			
	11	29/09/2017			
	11	30/10/2017			
	11	27/11/2017			
	11	07/12/2017			
	11	21/05/2018			
	11	28/06/2018			
	11	24/07/2018			
	11	25/08/2018			
	11	25/09/2018			
	11	23/10/2018			
	11	22/11/2018			
	11	08/12/2018			
	11	21/05/2019			
	11	20/06/2019			
9	12	04/07/17	Trafo 2 ESP	3	
	12	30/07/2017			
	12	30/08/2017			
	12	29/09/2017			
	12	30/10/2017			



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium															
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612						
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida		
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	
	12	27/11/2017						0,017		25	0,5		C		28	B							
	12	07/12/2017						0,022	29,7	29	0,1	0,8106	C		44	D	42	19.523	66.059	3	102	912	
	12	21/05/2018						0,022		28	0,5		C		30	E							
	12	28/06/2018						0,025	30,2	34	0,5	0,8080	C		16	C	43	22.272	68.719	4	38	1.226	
	12	24/07/2018						0,022		30	0,5		C		25	D							
	12	25/08/2018						0,022		67	0,5		C		11	C							
	12	25/09/2018						0,022		31	0,5		C		32	D							
	12	23/10/2018						0,025		54	0,5		C		13	B							
	12	22/11/2018						0,025		33	0,5		C		38	D							
	12	08/12/2018						0,025		30	0,5		C		32	C							
	12	21/05/2019						0,028		40	1,0		C		18	C							
	12	20/06/2019						0,028	31,3	41	1,0	0,8163	C		18	D	13	21.221	65.168	7	48	1.870	
10	13	04/07/17	Trafo 3 ESP	3	007208428	2010	731	Zhejiang	0,025	27,8	40	1,5	0,8415	C		27	D	21	16.364	70.694	3	38	1.068
	13	30/07/2017						0,025		66	1,5		C		18	D							
	13	30/08/2017						0,025		50	1,5		C		21	C							
	13	29/09/2017						0,028		59	1,5		C		22	C							
	13	30/10/2017						0,028		55	1,5		C		28	D							
	13	27/11/2017						0,014		39	0,5		C		24	B							
	13	07/12/2017						0,014	39,8	45	0,5	0,8099	C		29	B	15	14.515	65.277	2	40	559	
	13	21/05/2018						0,014		54	0,5		C		23	D							
	13	28/06/2018						0,017	38,4	43	0,5	0,8075	C		22	C	30	15.202	64.188	4	65	1.833	
	13	24/07/2018						0,014		43	0,5		C		20	C							
	13	25/08/2018						0,014		76	0,5		C		11	C							
	13	25/09/2018						0,020		47	0,5		C		20	B							
	13	23/10/2018						0,014		49	0,5		C		15	A							
	13	22/11/2018						0,017		39	0,5		C		15	A							
	13	08/12/2018						0,014		34	0,5		C		18	A							
	13	21/05/2019						0,014		41	0,5		C		20	C							
	13	20/06/2019						0,020	38,5	36	0,5	0,8215	C		23	C	14	18.302	64.811	6	86	2.181	
11	14	04/07/17	Trafo 4 ESP	3	007108428	2010	731	Zhejiang	0,020	31,2	62	1,0	0,8420	C		19	D	10	18.669	66.036	3	24	665
	14	30/07/2017						0,022		60	1,0		C		15	C							
	14	30/08/2017						0,020		55	1,0		C		17	D							
	14	29/09/2017						0,022		59	1,0		C		15	C							
	14	30/10/2017						0,025		55	1,0		C		15	D							
	14	27/11/2017						0,022		69	1,0		C		16	C							
	14	07/12/2017						0,025	29,8	43	1,0	0,8471	C		19	D	14	15.414	63.712	2	22	675	
	14	21/05/2018						0,028		37	1,0		C		21	D							
	14	28/06/2018						0,025	30,5	71	1,0	0,8442	C		11	A	20	14.690	67.822	2	22	683	
	14	24/07/2018						0,022		42	1,0		C		11	C							
	14	25/08/2018						0,025		66	1,5		C		11	B							
	14	25/09/2018						0,022		51	1,5		C		11	B							
	14	23/10/2018						0,022		56	1,5		C		12	B							
	14	22/11/2018						0,022		44	1,5		C		11	B							
	14	08/12/2018						0,022		49	1,0		C		13	C							
	14	21/05/2019						0,022		58	1,0		C		11	C							



no	Test Cer.	Date Test	nama unit	m cennix											corr. sul	keterangan	MALAYSIAN LAB RECOMMENDATION	
								lpf (C)		dbpc	furan	metal in oil						
				C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu				
Standa				ethane	ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B			
					< 65	<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0,00	0,00	0,00		Non corrosive	
	12	27/11/2017																Perform OIL REGENERATION
	12	07/12/2017			0	0	0	147	0,000	0,030	0,20	0	0,00	0,00	0,00	3A	Non corrosive	Perform OIL REGENERATION
	12	21/05/2018																Perform OIL PURIFICATION
	12	28/06/2018			0	0	0	85	0,000	0,040	0,19	0	0,00	0,00	0,02	3 A	Suspec Corrosive	Retest 12 months: Specific Gravity Add Passivator
	12	24/07/2018																Perform OIL REGENERATION
	12	25/08/2018																Periodic condition monitoring (Purifying on end of July 2018)
	12	25/09/2018																Perform OIL REGENERATION
	12	23/10/2018																Periodic condition monitoring
	12	22/11/2018																Perform for Purification
	12	08/12/2018																Perform for Purification
	12	21/05/2019																Periodic condition monitoring
	12	20/06/2019			0	0	0	68	0,000	0,010	0,19	180	0,00	0,00	0,02	3A	Non corrosive	Periodic condition monitoring
10	13	04/07/17	Trafo 3 ESP	3	0	0	0	62	0,000	0,620	0,01	0	0,00	0,01	0,00	2 C	Non corrosive	Perform OIL REGENERATION
	13	30/07/2017																Perform OIL REGENERATION
	13	30/08/2017																Perform OIL REGENERATION
	13	29/09/2017																Perform OIL REGENERATION
	13	30/10/2017																Perform OIL REGENERATION
	13	27/11/2017																Perform OIL REGENERATION
	13	07/12/2017			0	0	0	57	0,000	.08	.21	0	0,00	0,00	0,00	2E	Non corrosive	Perform OIL PURIFICATION
	13	21/05/2018																Perform OIL PURIFICATION
	13	28/06/2018			0	0	0	99	0,000	0,010	0,20	4	0,00	0,00	0,02	2 E	Non corrosive	Retest 12 months: Specific Gravity Retest 6 months: Dielectric and Water Content Add Passivator
	13	24/07/2018																Perform OIL PURIFICATION
	13	25/08/2018																Periodic condition monitoring (Purifying on end of July 2018)
	13	25/09/2018																Perform OIL PURIFICATION
	13	23/10/2018																Periodic condition monitoring
	13	22/11/2018																Periodic condition monitoring
	13	08/12/2018																Periodic condition monitoring
	13	21/05/2019																ReTest in 6 months : Water Content and BDV - Prepare for PURIFICATION
	13	20/06/2019			0	0	0	106	0,000	0,070	0,19	0	0,00	0,00	0,02	2E	Non corrosive	ReTest in 6 months : Water Content and BDV - Prepare for PURIFICATION
11	14	04/07/17	Trafo 4 ESP	3	0	0	0	37	0,000	1,360	0,01	0	0,00	0,00	0,00	3 A	Suspec Corrosive	Retest 6 months: Acidity, IFT and LPF Add inhibitor 2,6-DBPC and Passivator Other parameters: Periodic Condition Monitoring
	14	30/07/2017																Retest 6 months: Acidity, IFT and Liquid Power Factor Other parameters: Periodic Condition Monitoring
	14	30/08/2017																Periodic condition monitoring
	14	29/09/2017																Periodic condition monitoring
	14	30/10/2017																Periodic condition monitoring
	14	27/11/2017																Periodic condition monitoring
	14	07/12/2017			0	0	0	38	0,000	0,370	0,03	0	0,00	0,00	0,00	1B	Non corrosive	Retest 6 months: Acidity, IFT
	14	21/05/2018																Retest 6 months: Dielectric and Water Content
	14	28/06/2018			0	0	0	44	0,000	0,550	0,00	0	0,00	0,03	0,02	1 B	Non corrosive	Add inhibitor 2,6-DBPC and PASSIVATOR
	14	24/07/2018																Periodic condition monitoring
	14	25/08/2018																Periodic condition monitoring (Purifying on end of July 2018)
	14	25/09/2018																Periodic condition monitoring
	14	23/10/2018																Periodic condition monitoring
	14	22/11/2018																Periodic condition monitoring
	14	08/12/2018																Periodic condition monitoring
	14	21/05/2019																Periodic condition monitoring



no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION	
Standar					
	12	27/11/2017			
	12	07/12/2017			
	12	21/05/2018			
	12	28/06/2018			
	12	24/07/2018			
	12	25/08/2018			
	12	25/09/2018			
	12	23/10/2018			
	12	22/11/2018			
	12	08/12/2018			
	12	21/05/2019			
	12	20/06/2019			
10	13	04/07/17	Trafo 3 ESP	3	
	13	30/07/2017			
	13	30/08/2017			
	13	29/09/2017			
	13	30/10/2017			
	13	27/11/2017			
	13	07/12/2017			
	13	21/05/2018			
	13	28/06/2018			
	13	24/07/2018			
	13	25/08/2018			
	13	25/09/2018			
	13	23/10/2018			
	13	22/11/2018			
	13	08/12/2018			
	13	21/05/2019			
	13	20/06/2019			
11	14	04/07/17	Trafo 4 ESP	3	
	14	30/07/2017			
	14	30/08/2017			
	14	29/09/2017			
	14	30/10/2017			
	14	27/11/2017			
	14	07/12/2017			
	14	21/05/2018			
	14	28/06/2018			
	14	24/07/2018			
	14	25/08/2018			
	14	25/09/2018			
	14	23/10/2018			
	14	22/11/2018			
	14	08/12/2018			
	14	21/05/2019			



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium															
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612						
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida		
		BATASAN OPERASI						< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	
	14	20/06/2019						0,022	30,3	51	1,0	0,8451	C		12	C	19	16.312	66.417	4	37	732	
12	15	04/07/17	Ash Handling		10820108	2011	1.023	Wolong	0,022	38,1	38	1,0	0,8779	C		28	C	35	10.984	71.058	20	441	5.037
	15	30/07/2017							0,025		56	1,0		C		21	D						
	15	30/08/2017							0,022		53	1,0		C		16	B						
	15	29/09/2017							0,022		57	1,0		C		16	B						
	15	30/10/2017							0,022		45	1,0		C		20	B						
	15	27/11/2017							0,020		36	1,0		C		29	C						
	15	07/12/2017							0,017	37,5	34	1,0	0,8932	C		26	B	49	12.034	67.124	20	558	5.440
	15	21/05/2018							0,022		33	1,0		C		26	C						
	15	28/06/2018							0,025	37,5	42	1,0	0,8804	C		24	C	53	10.939	69.394	20	557	5.425
	15	24/07/2018							0,022		39	1,0		C		20	B						
	15	25/08/2018							0,025		60	1,5		C		16	B						
	15	25/09/2018							0,025		43	1,5		C		17	A						
	15	23/10/2018							0,022		28	1,5		C		19	C						
	15	22/11/2018							0,022		24	1,5		C		28	B						
	15	08/12/2018							0,022		20	1,5		C		28	B						
	15	21/05/2019							0,022		28	1,5		C		29	C						
	15	20/06/2019							0,022	36,6	27	1,5	0,8890	C		29	C	53	14.032	65.585	13	487	5.360
13	16	04/07/17	Ash Handling		10820107	2011	1.023	Wolong	0,022	37,3	36	1,0	0,8788	C		21	C	18	21.157	64.224	4	129	1.676
	16	30/07/2017							0,022		40	1,0		C		32	D						
	16	30/08/2017							0,022		31	1,0		C		29	C						
	16	29/09/2017							0,022		28	1,5		C		31	B						
	16	30/10/2017							0,020		46	1,5		C		21	C						
	16	27/11/2017							0,022		47	1,5		C		20	C						
	16	07/12/2017							0,020	36,6	24	1,5	0,8940	C		32	D	20	15.791	63.530	5	275	2.783
	16	21/05/2018							0,020		24	1,5		C		32	D						
	16	28/06/2018							0,022	35,0	50	1,5	0,8811	C		16	B	32	22.904	67.139	4	249	3.112
	16	24/07/2018							0,020		27	1,5		C		30	C						
	16	25/08/2018							0,022		45	1,5		C		16	B						
	16	25/09/2018							0,022		36	1,5		C		19	B						
	16	23/10/2018							0,022		36	1,5		C		23	B						
	16	22/11/2018							0,022		30	1,5		C		25	B						
	16	08/12/2018							0,022		32	1,5		C		24	B						
	16	21/05/2019							0,022		32	1,5		C		23	B						
	16	20/06/2019							0,022	36,8	37	1,5	0,8889	C		24	B	39	11.205	66.397	21	439	5.998
14	17	27/11/17	Trafo 1 ESP	2	10037	1010	731	Fujian	0,020		48	1,0		C		21	D						
	17	07/12/17							0,022		22	1,0		C		35	E						
	17	21/05/2018							0,025		24	1,0		C		32	E						
	17	28/06/2018							0,025	32,9	77	1,0	0,8693	C		11	B	93	17.225	43.231	5	57	775
	17	24/07/2018							0,028		69	1,0		C		11	B						
	17	25/08/2018							0,031		74	1,5		C		12	B						
	17	25/09/2018							0,028		74	1,5		C		10	B						
	17	23/10/2018							0,028		54	1,5		C		11	B						
	17	22/11/2018							0,028		54	1,5		C		10	B						
	17	08/12/2018							0,028		46	1,5		C		10	B						
	17	21/05/2019							0,031		67	1,5		C		12	B						
	17	20/06/2019							0,031	31,0	62	1,0	0,8776	C		13	C	62	18.560	46.166	5	50	839



no	Test Cer.	Date Test	nama unit	m cennix											corr. sul	keterangan	MALAYSIAN LAB RECOMMENDATION	
								lpf (C)		dbpc	furan	metal in oil						
				C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	TOT	25°	100°			Al	Fe	Cu				
Standa				ethane	ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B			
					< 65	<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0,00	0,00	0,00		Non corrosive	
	14	20/06/2019			0	0	0	60	0,000	0,490	0,00	0	0,00	0,03	0,01	1B	Non corrosive	Periodic condition monitoring
12	15	04/07/17	Ash Handling		7	8	0	511	0,000	0,730	0,23	0	0,00	0,01	0,01	1 B	Non corrosive	Add Passivator Periodic Condition Monitoring
	15	30/07/2017																Periodic condition monitoring
	15	30/08/2017																Periodic condition monitoring
	15	29/09/2017																Periodic condition monitoring
	15	30/10/2017																Periodic condition monitoring
	15	27/11/2017																Periodic condition monitoring
	15	07/12/2017			8	9	0	644	0,000	0,240	0,24	0	0,00	0,00	0,01	2C	Non corrosive	Periodic condition monitoring
	15	21/05/2018																Periodic condition monitoring
	15	28/06/2018			9	8	0	647	0,000	0,460	0,22	0	0,00	0,01	0,04	1 B	Non corrosive	Add passivator
	15	24/07/2018																Periodic condition monitoring
	15	25/08/2018																Periodic condition monitoring
	15	25/09/2018																Periodic condition monitoring
	15	23/10/2018																Periodic condition monitoring
	15	22/11/2018																Monitoring : Water Content and BDV - Prepare for PURIFICATION
	15	08/12/2018																Retest in 6 months : Water Content and BDV - Prepare for PURIFICATION
	15	21/05/2019																Perform for PURIFICATION
	15	20/06/2019			0	11	0	564	0,000	0,360	0,21	0	0,00	0,00	0,02	2e	Non corrosive	Perform for PURIFICATION
13	16	04/07/17	Ash Handling		0	1	0	152	0,000	1,350	0,22	0	0,00	0,01	0,00	3 A	Suspec Corrosive	Add Passivator Periodic Condition Monitoring
	16	30/07/2017																Retest 6 months: Dielectric and Water Content Other parameters: Periodic Condition Monitoring
	16	30/08/2017																Periodic Condition Monitoring
	16	29/09/2017																Retest 6 months: Dielectric and Water Content
	16	30/10/2017																Retest 6 months: Dielectric and Water Content
	16	27/11/2017																Periodic Condition Monitoring
	16	07/12/2017			0	4	0	304	0,000	0,270	0,24	3	0,00	0,00	0,01	1A	Non corrosive	Perform OIL PURIFICATION
	16	21/05/2018																Perform OIL PURIFICATION
	16	28/06/2018			0	5	0	290	0,000	0,490	0,20	0	0,00	0,00	0,02	3 A	Suspec Corrosive	Retest 6 months: DGA Add Passivator
	16	24/07/2018																Perform OIL PURIFICATION
	16	25/08/2018																Periodic condition monitoring (Purifying on end of July 2018)
	16	25/09/2018																Periodic condition monitoring
	16	23/10/2018																Periodic condition monitoring
	16	22/11/2018																Periodic condition monitoring
	16	08/12/2018																Periodic condition monitoring
	16	21/05/2019																Periodic condition monitoring
	16	20/06/2019			11	9	0	519	0,000	0,680	0,20	0	0,01	0,00	0,02	3A	Non corrosive	Periodic condition monitoring
14	17	27/11/17	Trafo 1 ESP	2														Retest 6 months: Dielectric and Water Content Other parameters: Periodic Condition Monitoring
	17	07/12/17																OIL PURIFICATION
	17	21/05/2018																OIL PURIFICATION
	17	28/06/2018			0	2	0	157	0,030	3,420	0,00	0	0,00	0,06	0,03	2 C	Non corrosive	Add inhibitor 2,6-DBPC and PASSIVATOR
	17	24/07/2018																Periodic condition monitoring
	17	25/08/2018																Periodic condition monitoring
	17	25/09/2018																Periodic condition monitoring
	17	23/10/2018																Periodic condition monitoring
	17	22/11/2018																Periodic condition monitoring
	17	08/12/2018																Periodic condition monitoring
	17	21/05/2019																Periodic condition monitoring
	17	20/06/2019			0	0	0	117	0,000	0,720	0,00	0	0,00	0,00	0,02	3A	Non corrosive	Monitoring : Acidity, IFT and LPF



no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION	
Standar					
	14	20/06/2019			
12	15	04/07/17	Ash Handling		
	15	30/07/2017			
	15	30/08/2017			
	15	29/09/2017			
	15	30/10/2017			
	15	27/11/2017			
	15	07/12/2017			
	15	21/05/2018			
	15	28/06/2018			
	15	24/07/2018			
	15	25/08/2018			
	15	25/09/2018			
	15	23/10/2018			
	15	22/11/2018			
	15	08/12/2018			
	15	21/05/2019			
	15	20/06/2019			
13	16	04/07/17	Ash Handling		
	16	30/07/2017			
	16	30/08/2017			
	16	29/09/2017			
	16	30/10/2017			
	16	27/11/2017			
	16	07/12/2017			
	16	21/05/2018			
	16	28/06/2018			
	16	24/07/2018			
	16	25/08/2018			
	16	25/09/2018			
	16	23/10/2018			
	16	22/11/2018			
	16	08/12/2018			
	16	21/05/2019			
	16	20/06/2019			
14	17	27/11/17	Trafo 1 ESP	2	
	17	07/12/17			
	17	21/05/2018			
	17	28/06/2018			
	17	24/07/2018			
	17	25/08/2018			
	17	25/09/2018			
	17	23/10/2018			
	17	22/11/2018			
	17	08/12/2018			
	17	21/05/2019			
	17	20/06/2019			



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium														
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612					
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida	
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500
15	18	27/11/17	Trafo 2 ESP	2	10036	1010	731 Fujian	0,048		55	1,0		C		13	C						
	18	07/12/17						0,051		59	1,0		C		14	C						
	18	21/05/2018						0,053		55	1,0		C		16	C						
	18	28/06/2018						0,056	31,3	66	1,0	0,8573	C		13	C	8	22.714	64.906	5	67	879
	18	24/07/2018						0,059		79	1,5		C		8	A						
	18	25/08/2018						0,062		67	1,5		C		10	B						
	18	25/09/2018						0,056		51	1,5		C		9	B						
	18	23/10/2018						0,062		67	1,5		C		13	B						
	18	22/11/2018						0,062		61	1,5		C		8	A						
	18	08/12/2018						0,062		59	1,5		C		8	A						
	18	21/05/2019						0,065		77	1,0		C		11	B						
	18	20/06/2019						0,065	29,7	76	1,0	0,8655	C		12	B	5	29.439	65.338	5	25	977
16	19	27/11/17	Trafo 3 ESP	2	10033	1010	731 Fujian	0,053		42	1,5		C		15	D						
	19	07/12/17						0,056		60	1,5		C		16	D						
	19	21/05/2018						0,059		51	1,5		C		18	D						
	19	28/06/2018						0,062	26,3	78	1,5	0,8602	C		14	C	14	17.627	48.672	4	47	935
	19	24/07/2018						0,070		67	1,5		C		12	B						
	19	25/08/2018						0,067		73	1,5		C		13	C						
	19	25/09/2018						0,067		59	1,5		C		12	B						
	19	23/10/2018						0,070		86	1,5		C		13	B						
	19	22/11/2018						0,070		13	1,5		C		84	E						
	19	08/12/2018						0,070		8	1,5		C		84	E						
	19	21/05/2019						0,073		50	1,5		C		13	C						
	19	20/06/2019						0,073	26,5	53	1,5	0,8687	C		13	C	16	17.449	51.923	5	58	1.049
17	20	27/11/17	Trafo 4 ESP	2	10032	1010	731 Fujian	0,020		62	1,0		C		16	D						
	20	07/12/17						0,022		58	1,0		C		17	D						
	20	21/05/2018						0,025		40	1,0		C		18	D						
	20	28/06/2018						0,028	32,6	88	1,0	0,8648	C		15	D	22	25.201	74.464	5	85	823
	20	24/07/2018						0,031		59	1,0		C		13	B						
	20	25/08/2018						0,031		68	1,5		C		10	B						
	20	25/09/2018						0,031		51	1,5		C		11	B						
	20	23/10/2018						0,034		57	1,0		C		15	C						
	20	22/11/2018						0,034		69	1,0		C		14	B						
	20	08/12/2018						0,034		61	1,0		C		18	D						
	20	21/05/2019						0,036		60	1,0		C		14	B						
	20	20/06/2019						0,036	30,3	61	1,0	0,8732	C		14	C	33	25.208	70.145	4	35	853
18	21	27/11/17	Main Trafo	2	1006902	2011	11.160 Wolong	0,008		70	0,5		C		16	B						
	21	07/12/17						0,008		67	0,5		C		13	A						
	21	21/05/2018						0,011		46	0,5		C		14	B						
	21	28/06/2018						0,008	42,1	75	0,5	0,8663	C		10	B	19	12.733	53.386	10	193	2.448
	21	24/07/2018						0,011		79	0,5		C		11	B						
	21	25/08/2018						0,011		83	0,5		C		9	A						
	21	25/09/2018						0,011		90	0,5		C		7	A						
	21	23/10/2018						0,011		64	0,5		C		7	A						
	21	22/11/2018						0,011		61	0,5		C		7	A						
	21	08/12/2018						0,011		57	0,5		C		6	A						
	21	21/05/2019						0,011		63	0,5		C		7	A						
	21	20/06/2019						0,011	39,3	58	0,5	0,8745	C		11	A	10	17.382	50.681	8	107	712



no	Test Cer.	Date Test	nama unit	m cennix											corr. sul	keterangan	MALAYSIAN LAB RECOMMENDATION	
								lpf (C)		dbpc	furan	metal in oil						
				C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	TOT	25°	100°			Al	Fe	Cu				
Standa				ethane	ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B			
					< 65	<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0,00	0,00	0,00		Non corrosive	
15	18	27/11/17	Trafo 2 ESP	2														Periodic condition monitoring
	18	07/12/17																Retest 6 months: Acidity and IFT
																		Periodic condition monitoring
	18	21/05/2018																Retest 6 months: Acidity and IFT
	18	28/06/2018			0	0	0	80	0,000	0,680	0,00	42	0,00	0,00	0,02	3 A	Suspec Corrosive	Add inhibitor 2,6-DBPC and PASSIVATOR
	18	24/07/2018																Periodic condition monitoring
	18	25/08/2018																Periodic condition monitoring
	18	25/09/2018																Perform OIL REGENERATION
	18	23/10/2018																Perform OIL REGENERATION
	18	22/11/2018																Perform OIL REGENERATION
	18	08/12/2018																Perform OIL REGENERATION
	18	21/05/2019																Perform OIL REGENERATION
	18	20/06/2019			0	0	0	35	0,000	0,450	0,00	0	0,00	0,00	0,02	3A	Non corrosive	Perform OIL REGENERATION
16	19	27/11/17	Trafo 3 ESP	2														Retest 6 months: Acidity and IFT
																		Other parameters: Periodic Condition Monitoring
	19	07/12/17																Retest 6 months: Acidity and IFT
																		Other parameters: Periodic Condition Monitoring
	19	21/05/2018																Retest 6 months: Acidity and IFT
	19	28/06/2018			0	0	0	65	0,000	0,570	0,00	0	0,00	0,00	0,02	3 A	Suspec Corrosive	Perform OIL REGENERATION
	19	24/07/2018																Perform OIL REGENERATION
	19	25/08/2018																Perform OIL REGENERATION
	19	25/09/2018																Perform OIL REGENERATION
	19	23/10/2018																Perform OIL REGENERATION
	19	22/11/2018																Perform OIL REGENERATION
	19	08/12/2018																Perform OIL REGENERATION
	19	21/05/2019																Perform OIL REGENERATION
	19	20/06/2019			0	0	0	79	0,010	0,790	0,00	0	0,00	0,00	0,02	3A	Non corrosive	Perform OIL REGENERATION
17	20	27/11/17	Trafo 4 ESP	2														Periodic condition monitoring
	20	07/12/17																Periodic condition monitoring
	20	21/05/2018																Periodic condition monitoring
	20	28/06/2018			0	0	0	112	0,000	1,620	0,01	0	0,00	0,00	0,03	2 C	Non corrosive	Add inhibitor 2,6-DBPC and PASSIVATOR
	20	24/07/2018																Periodic condition monitoring
	20	25/08/2018																Periodic condition monitoring
	20	25/09/2018																Periodic condition monitoring
	20	23/10/2018																Periodic condition monitoring
	20	22/11/2018																Periodic condition monitoring
	20	08/12/2018																Periodic condition monitoring
	20	21/05/2019																Periodic condition monitoring
	20	20/06/2019			0	0	0	72	0,020	1,150	0,01	7	0,00	0,00	0,02	2C	Non corrosive	Monitoring : Acidity, IFT and LPF
18	21	27/11/17	Main Trafo	2														Periodic condition monitoring
	21	07/12/17																Periodic condition monitoring
	21	21/05/2018																Periodic condition monitoring
	21	28/06/2018			0	2	0	224	0,000	0,410	0,22	113	0,00	0,00	0,03	1 B	Non corrosive	Retest 6 months: Furan and DGA Add Passivator
	21	24/07/2018																Periodic Condition monitoring
	21	25/08/2018																Periodic condition monitoring
	21	25/09/2018																Periodic condition monitoring
	21	23/10/2018																Periodic condition monitoring
	21	22/11/2018																Periodic condition monitoring
	21	08/12/2018																Periodic condition monitoring
	21	21/05/2019																Periodic condition monitoring
	21	20/06/2019			0	2	0	127	0,010	0,730	0,22	0	0,00	0,00	0,02	2E	Non corrosive	Periodic condition monitoring



no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION	
Standar					
15	18	27/11/17	Trafo 2 ESP	2	
	18	07/12/17			
	18	21/05/2018			
	18	28/06/2018			
	18	24/07/2018			
	18	25/08/2018			
	18	25/09/2018			
	18	23/10/2018			
	18	22/11/2018			
	18	08/12/2018			
	18	21/05/2019			
	18	20/06/2019			
16	19	27/11/17	Trafo 3 ESP	2	
	19	07/12/17			
	19	21/05/2018			
	19	28/06/2018			
	19	24/07/2018			
	19	25/08/2018			
	19	25/09/2018			
	19	23/10/2018			
	19	22/11/2018			
	19	08/12/2018			
	19	21/05/2019			
	19	20/06/2019			
17	20	27/11/17	Trafo 4 ESP	2	
	20	07/12/17			
	20	21/05/2018			
	20	28/06/2018			
	20	24/07/2018			
	20	25/08/2018			
	20	25/09/2018			
	20	23/10/2018			
	20	22/11/2018			
	20	08/12/2018			
	20	21/05/2019			
	20	20/06/2019			
18	21	27/11/17	Main Trafo	2	
	21	07/12/17			
	21	21/05/2018			
	21	28/06/2018			
	21	24/07/2018			
	21	25/08/2018			
	21	25/09/2018			
	21	23/10/2018			
	21	22/11/2018			
	21	08/12/2018			
	21	21/05/2019			
	21	20/06/2019			